

MEMORY RETRIEVAL UNDER RESTRICTED CONTEXT: AN ACCURACY AND COST STUDY

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ABSTRACT

This study compares file-based project memory against hybrid retrieval from working memory under equal context-character budgets. We evaluate answer accuracy and total model cost, including memory writes, using an adapted slice of LongMemEval-S. The baseline uses a truncated markdown file, while the enhanced method employs lexical Jaccard similarity with temporal decay for selection. Results indicate that hybrid retrieval achieves higher accuracy but incurs greater token costs due to write operations. At restricted budgets, truncation disadvantages the baseline. At expanded budgets, partial facts in the baseline can still yield correct answers despite lower evidence recall. These findings highlight tradeoffs between retrieval precision and computational overhead in constrained contexts.

1 INTRODUCTION

Managing long-term conversational history within fixed context windows remains a critical challenge for large language models. This research investigates how two distinct memory architectures perform when constrained by identical character budgets. Specifically, we compare a file-based project memory system against a hybrid retrieval mechanism operating on process-local working memory. The primary objective is to determine which approach yields superior answer accuracy and what associated costs arise in terms of total token usage, including both query processing and memory writing operations. By isolating these variables, we aim to clarify the practical implications of memory management strategies in resource-constrained environments.

2 RELATED WORK

Prior work has explored various methods for extending LLM context capabilities. MemGPT proposes an operating-system-like hierarchy for managing information beyond standard context limits, focusing on virtual context management (Packer et al., 2023). While this conceptual framework motivates hierarchical storage, our study does not implement or test its full architecture. Instead, we focus on simpler retrieval mechanisms. LongMemEval provides a benchmark for evaluating long-term interactive memory, covering extraction, reasoning, and updating tasks (Wu et al., 2025). Our experiment adapts a narrow subset of this benchmark, specifically single-session user slices, rather than claiming coverage of the entire evaluation suite. This distinction ensures that our results reflect specific architectural comparisons rather than general benchmark performance.

3 METHOD

We implemented two distinct memory rails to address the research question. Both arms utilize the same underlying model, system prompts, and fact sets to ensure comparability. The experimental design emphasizes isolation and precise cost accounting to distinguish between implementation details and tested behaviors. We carefully control for context length and tool availability to isolate the effects of the memory retrieval strategy itself.

3.1 FILE-BASED PROJECT MEMORY

The baseline arm employs ProjectMemoryRail, where facts are written to an isolated markdown file named JIUWENSWARM.md. During inference, the prompt includes file metadata and is truncated to fit the full-section character budget. This approach relies on static file access rather than dynamic retrieval. It is important to note that this is not a retrieval-augmented embedding baseline; it simply loads available text up to the limit. Both arms receive identical facts and questions without external tools, ensuring that differences arise solely from the memory handling mechanism.

3.2 HYBRID SELECTION FROM WORKING MEMORY

The enhanced arm utilizes EnhancedMemoryRail, which writes completed exchanges to process-local L1 memory. Selection relies on lexical Jaccard similarity combined with temporal decay, access frequency, and importance scores, avoiding learned embeddings. Selected memories are assembled under the same character budget as the baseline. Although the broader architecture supports L2 and L3 classes, this specific rail does not route writes there or persist data across restarts. MemGPT serves as conceptual context for hierarchical memory but is not implemented or tested as a baseline in this experiment (Packer et al., 2023).

3.3 ISOLATION AND COST ACCOUNTING

Each public-data question is paired with a unique fact episode and fresh agents to prevent cross-contamination. The baseline writes facts to disk, while the enhanced arm performs model-backed memory-writing conversations. A new conversation ID is used for each formal question. The runner verifies isolated query history and actual memory injection, alternating arm order to mitigate position bias. Total cost records both query usage and memory-write usage. Latency including writes is amortized over paired observations. Correctness is determined by accepted-answer string matching, while evidence recall requires the complete target fact to survive prompt truncation (Wu et al., 2025).

4 EXPERIMENTAL SETUP

We conducted experiments using an adapted slice of LongMemEval-S, converting histories into fixed fact episodes with target answer turns and distractors. Two context budgets were tested: restricted and expanded. Each setting involved paired observations to compare accuracy and cost metrics directly. The following subsections detail the specific configurations for each budget level.

4.1 RESTRICTED CONTEXT BUDGET

In the restricted context budget setup, the character limit was set to 600. This configuration included 20 unique questions, resulting in 20 paired observations with 1 repeat. Metrics tracked included answer accuracy, complete evidence recall, total tokens per question including writes, and latency measurements. This tight constraint was designed to stress-test the truncation behavior of the baseline versus the selective retrieval of the enhanced arm.

4.2 EXPANDED CONTEXT BUDGET

The expanded context budget setup increased the character limit to 2400. This condition also featured 20 unique questions, yielding 20 paired observations with 1 repeat. Similar to the small budget, we measured accuracy, evidence recall, total token costs, and latency. The larger window allowed more content to be included in the baseline prompt, potentially reducing truncation penalties and altering the relative advantage of selective retrieval.

5 RESULTS AND ANALYSIS

We analyzed the performance of both memory rails across the two budget conditions. The results reveal distinct patterns in accuracy, evidence retention, and computational cost. Below, we detail

the findings for the restricted and expanded context scenarios, highlighting the tradeoffs observed in each case.

5.1 RESTRICTED CONTEXT BUDGET

Under the restricted budget, the baseline achieved an accuracy of 0.0% with an evidence recall of 0.0%. In contrast, the enhanced arm reached an accuracy of 65.0% and an evidence recall of 65.0%, representing an accuracy gain of 65 percentage points. However, this improvement came at a higher cost: the enhanced arm consumed 5083.45 tokens per question compared to 234.30 for the baseline. Query latency was 503.76 ms for the enhanced arm versus 408.03 ms for the baseline, while end-to-end latency was 7375.67 ms and 408.03 ms respectively. The baseline’s lower accuracy is partly attributed to truncation and metadata overhead consuming the limited budget.

5.2 EXPANDED CONTEXT BUDGET

With the expanded budget, the baseline recorded an accuracy of 60.0% and evidence recall of 55.0%. The enhanced arm achieved an accuracy of 95.0% and evidence recall of 95.0%, with an accuracy gain of 35 percentage points. Notably, baseline answer accuracy exceeded complete-evidence recall because partial facts could still contain the necessary answer components. Cost-wise, the enhanced arm used 13201.80 tokens per question versus 843.85 for the baseline. Interestingly, query latency decreased for the enhanced arm to 601.41 ms compared to 717.01 ms, though end-to-end latency remained higher at 7133.41 ms versus 717.01 ms due to write operations.

Table 1: Restricted Context Budget. Costs include memory writes; missing usage is not treated as zero.

Metric	Baseline	Enhanced
Correct answers	0	13
Paired observations	20	20
Answer accuracy	0.0%	65.0%
Evidence recall	0.0%	65.0%
Query tokens (total)	4686	3864
Write tokens (total)	0	97805
Total tokens / paired observation	234.30	5083.45
Mean query latency (ms)	408.03	503.76
Amortized total latency (ms)	408.03	7375.67
Excluded records	0	0

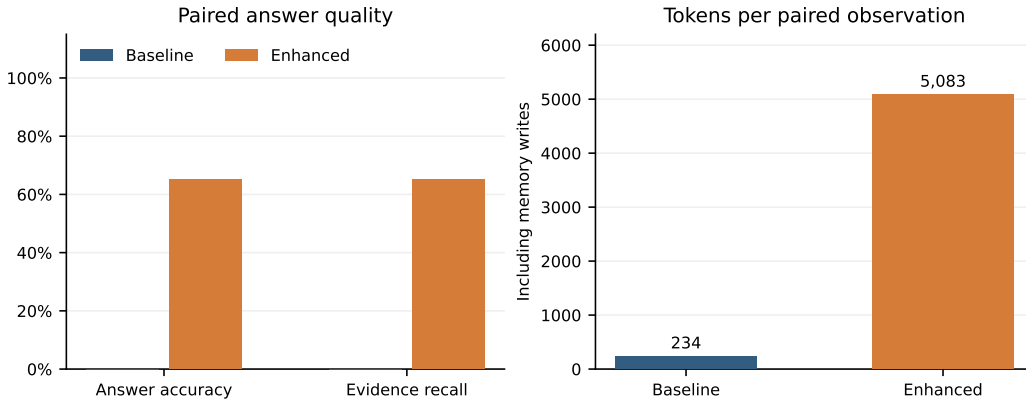


Figure 1: Restricted Context Budget. Quality and amortized total token cost on the same valid paired observations. No confidence interval is estimated.

Table 2: Expanded Context Budget. Costs include memory writes; missing usage is not treated as zero.

Metric	Baseline	Enhanced
Correct answers	12	19
Paired observations	20	20
Answer accuracy	60.0%	95.0%
Evidence recall	55.0%	95.0%
Query tokens (total)	16877	14638
Write tokens (total)	0	249398
Total tokens / paired observation	843.85	13201.80
Mean query latency (ms)	717.01	601.41
Amortized total latency (ms)	717.01	7133.41
Excluded records	0	0

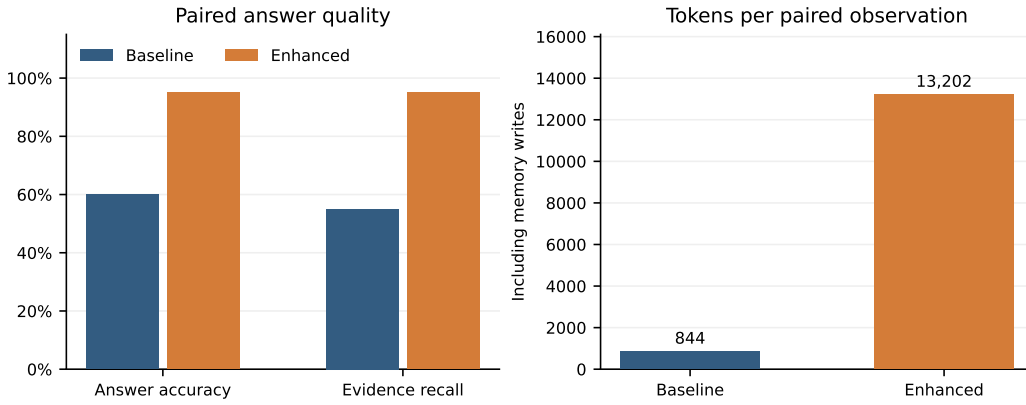


Figure 2: Expanded Context Budget. Quality and amortized total token cost on the same valid paired observations. No confidence interval is estimated.

6 DISCUSSION AND LIMITATIONS

These results illustrate the tension between retrieval precision and computational overhead. The enhanced hybrid retrieval consistently outperformed the file-based baseline in accuracy, particularly under restricted budgets where truncation severely impacted the baseline. However, this accuracy gain required additional token consumption for memory writes and selection processes. The discrepancy between answer accuracy and evidence recall in the expanded budget suggests that models can sometimes infer correct answers from partial context, complicating the interpretation of recall metrics. Several limitations must be acknowledged. First, this is a paired memory question-answering experiment, not an end-to-end paper-generation evaluation. Second, the current enhanced Rail only writes to process-local L1 memory; L2/L3 persistence and team synchronization remain untested. Third, string-based accepted-answer matching may miss semantic nuances, and context metadata consumes part of the character budget. Fourth, repeated questions sharing a dataset are not independent samples, so no significance tests are claimed. Finally, total token costs include all recorded attempts, and this adaptation of LongMemEval does not represent an official benchmark score.

7 CONCLUSION

In conclusion, hybrid retrieval from working memory offers higher answer accuracy than file-based project memory under equal context-character budgets, albeit at a higher total token cost due to memory write operations. The advantage of selective retrieval is most pronounced in restricted contexts where truncation penalizes static file loading. As budgets expand, the gap narrows, and baseline performance improves as partial facts become sufficient for answering. Future work should

explore persistent multi-level memory architectures and evaluate semantic scoring methods to better understand the tradeoffs between accuracy, cost, and latency in long-context applications.

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